Title: Integrative Experiments Identify How Punishment Impacts Welfare in Public Goods Games

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Abstract: Despite decades of research, the conditions under which punishment promotes cooperation remain unclear. Through an integrative experiment varying 14 design parameters across 360 experimental conditions (147,618 decisions from 7,100 participants), we reveal substantial heterogeneity in punishment effectiveness: its impact on welfare ranges from 43% improvement to 44% reduction depending on game parameters. To characterize these patterns, we developed models that outperformed human forecasters in predicting punishment effectiveness in new experiments. Communication emerges as the dominant factor, followed by contribution framing (opt-out vs. opt-in), contribution type (variable vs. all-or-nothing), game length, and outcome visibility, though these factors often interact. The results reframe the debate from whether punishment works to when it does, demonstrating how integrative experiments enable discovery of generalizable patterns in social phenomena.

Main Text:

Introduction

Human societies face innumerable situations where individual and collective interests conflict. These conflicts, known as social (or cooperative) dilemmas, arise in settings as diverse as sustainable resource management (*1*, *2*), climate change (*3–5*), tax compliance (*6*), and pandemic response (*7–9*). In these situations, cooperation—defined as choosing collective over individual benefit—becomes difficult to achieve and sustain. Public goods games (PGGs) provide a stylized framework capturing the essential features of real-world cooperation problems while remaining amenable to theoretical modeling and controlled experiments (*10*, *11*). In a typical PGG, individuals decide how much to contribute to a shared project that benefits everyone in the group. Contributions are multiplied by a factor (M), and the resulting amount is shared equally among all N group members; thus, everyone receives M/N per unit contributed to the project—often called the marginal per capita return ($MPCR$). When the $MPCR$ is less than 1 ($M < N$), individuals face the classic social dilemma: while the group as a whole benefits most when everyone contributes fully, individuals can maximize their personal gain by contributing nothing (Fig. 1.A). Failure to cooperate, where rational pursuit of self-interest undermines collective welfare, leads to the "free-rider problem" and the "tragedy of the commons" (*1*, *12*, *13*).

Costly peer punishment, which allows individuals to impose penalties on peers based on their actions in the PGG, has been widely studied over the past 25 years as a way to promote cooperation in social dilemmas. Early studies demonstrated that the ability to punish effectively deters free-riding and increases cooperation in terms of contributions to the public good (*14–18*). However, because punishment is costly for both the punisher and the punished, higher contributions to the public good do not necessarily translate to improved collective welfare. Several studies found that punishment's costs can outweigh its benefits in promoting cooperation, leading to reduced overall welfare (*19–21*). Subsequent research examined how various contextual factors influence whether increased cooperation outweighs punishment costs. These factors include structural design parameters like the number of interaction rounds (*22–24*), group size (*25, 26*), and payoff structure (*27*); the punishment cost and relative magnitude (*28, 29*); the existence of other mechanisms like reward (30); and social dynamics, including the ability to communicate (*31*), the type of peer information available (*32*), the benefits of reputation (*33, 34*), and the ability to coordinate against free-riders (*35, 36*).

Despite this large body of research, the specific conditions under which punishment improves group welfare remain unclear. We argue that this lack of clarity derives from the dominant experimental paradigm, which manipulates only one or a few theoretically informed factors in isolation (*37, 38*). For instance, a study might vary the time horizon while holding constant group size, payoff structure, punishment cost, and time pressure, assuming these to be inconsequential "nuisance" parameters. Because such studies differ in unspecified ways (experimental procedures, parameter values, participant populations, and "all else equal" assumptions), their results are difficult to compare or integrate. For example, one cannot say from these studies whether the number of rounds has a greater impact on punishment effectiveness than does the strength of the punishment mechanism, or whether the effect of punishment strength depends on game length, group size, and the existence of reward mechanisms. Even meta-analyses cannot fully address this issue. Setting aside publication bias (which may exclude null or negative results), the underlying studies were not designed to be comparable, so post-hoc integration is intrinsically challenging. As a consequence, a research program can establish a long list of factors that have some effect, but cannot say how much each matters or how they work together (*37, 39, 40*). Without knowing how these factors combine to determine outcomes, we cannot predict when punishment will help or harm welfare. We therefore ask: when does punishment improve versus undermine welfare, which factors matter most, and how do they interact?

Here, we conducted an integrative experiment (*37*) that systematically varied 14 parameters (Table 1) that describe the design space of PGGs, encompassing fundamental game structure (group size, game length, contribution mechanism), communication and information structure (availability of communication, visibility of peer actions and outcomes, knowledge of the game horizon), and incentive structure (MPCR, punishment and reward mechanisms). Each PGG scenario is represented as a point in this 14-dimensional space, allowing us to quantify similarities and differences between scenarios in terms of these potentially relevant attributes. We then systematically sampled experiments from this design space. Each experiment comprised two games—one with punishment enabled (treatment) and one without (control)—that otherwise shared identical parameter values.

We began with a set of learning experiments (Wave 1) in which we used a Sobol sequence, a quasi-random strategy that fills the space of possibilities evenly *(41–43)*, to select 320 experimental conditions. Each condition was sampled once to maximize breadth across the parameter space. We then conducted validation experiments (Wave 2), which served as a confirmatory phase. Using pre-registered conditions and sample sizes (see Methods), we selected 20 new, randomly chosen experiments (40 conditions), each with 8-12 trials. While Wave 1 prioritized breadth for pattern discovery, Wave 2 required higher precision for evaluating specific predictions with sufficient statistical power. This two-phase approach mirrors best practices in theory development by first identifying patterns and distilling them into a model, then testing whether that model's predictions and implications hold in completely new data, which directly addresses concerns about spurious patterns (i.e., overfitting) that might not generalize beyond the exploratory data (*48, 49*). Fig. 1.B illustrates the overall design.

Across both waves, we sampled 360 unique experimental conditions, yielding 147,618 decisions (102,343 contributions, 15,471 punishments, and 29,804 rewards) from 7,100 participants. Additional experiment design details are in the Methods section.

The integrative approach has several key advantages over single-factor studies and meta-analyses: (1) all factors were manipulated systematically and simultaneously, which allows us to measure relative importance and characterize interactions; (2) all experiments were run under similar protocols (e.g., participant recruitment, user interface), which enables direct comparisons and minimizes hidden moderators; (3) we avoided publication bias by treating all sampled conditions as informative regardless of statistical significance; (4) all inferences were evaluated on the pre-registered Wave 2 validation experiments, which guards against overfitting.

Results

Heterogeneous effects of punishment across experimental conditions

Punishment increased average contributions from 73% to 80% of endowment (coefficient = 0.066, SE = 0.018, 95% CI [0.029, 0.103], $p < 0.001$, N = 53,973 individual-level contribution decisions). This overall positive effect on contributions was replicated in the validation experiments where, on average, punishment increased contributions from 74% to 82% of endowment (coefficient = 0.077, SE = 0.016, 95% CI [0.045, 0.108], $p < 0.001$, N = 48,370 individual-level contribution decisions), and aligns with earlier studies that demonstrated punishment's ability to deter free-riding and increase contributions (*14–18*, *51*). For further details, see Methods.

However, as prior research has also emphasized, increased contributions do not necessarily translate to improved collective outcomes (i.e., overall earnings) once punishment costs are taken into account (*19–21*). To capture punishment's effect on overall earnings, we examine normalized efficiency, which scales group earnings relative to the minimum without punishment (where nobody contributes anything, "full defection") and the maximum without reward ("full cooperation") in each setting (see Methods for details). To illustrate, consider two PGGs with the same configuration (number of players/rounds, information availability, etc.) except that Game 1 has a multiplier of 2 and Game 2 has a multiplier of 4. A fully defecting group in Game 1 achieves a standard efficiency of 0.5, while a fully defecting group in Game 2 achieves 0.25, despite the groups exhibiting the same degree of cooperation. Normalized efficiency addresses this discrepancy by scaling each group's earnings relative to the natural benchmarks of full cooperation and defection.

Although punishment consistently increased contributions, its effect on welfare was complicated. On average, punishment reduced normalized efficiency from 0.71 to 0.63 (11% decrease) in the learning experiments (coefficient = -0.089, SE = 0.032, 95% CI [-0.151, -0.027], p = 0.005, N = 335 groups), with a smaller but directionally consistent decrease from 0.72 to 0.68 (6% decrease) in the validation experiments (coefficient = -0.043, SE = 0.022, 95% CI [-0.087, 0.001], p = 0.056, N = 417 groups). However, the overall average masks substantial and significant heterogeneity across PGG settings. As Figs. 2D and 2F show, observed punishment effect sizes varied from strongly positive to strongly negative in the learning experiments. Because each experimental condition in the learning experiments was sampled once, we clustered the observed conditions in the design space, putting similar conditions into 20 clusters using k-means to enable standard error estimation and visualization (see Methods). Statistical tests of heterogeneity (Cochran's Q-statistic and Fisherian permutation tests; see Methods) consistently indicated that this variation exceeded what would be expected by chance, both in the learning [Q(19) = 29.01, p = 0.066; permutation tests $p < 0.01$] and validation experiments [Q(19) = 30.29, p = 0.048; permutation tests $p < 0.01$], and when aggregated across both waves [Q(39) = 59.34, p = 0.019; permutation tests $p < 0.01$], with between-experimental-condition differences explaining a substantial portion of this variation (I^2 = 34% in learning; I^2 = 37% in validation; see Methods).

Because all experiments used identical protocols, recruitment, and interfaces, this heterogeneity can be more confidently attributed to game parameter differences, unlike meta-analyses where variation stems from differences in research teams, time periods, populations, experimental procedures, etc. (*54*, *55*). Therefore, the observed I^2 values represent a tighter measure of meaningful heterogeneity in the effect of punishment than is typically seen in meta-analyses. The magnitude of this heterogeneity is substantial. As shown in Figs. 2C and 2D, the learning experiments suggested that punishment could dramatically reduce normalized efficiency from 0.78 to 0.44 (44% decrease) in some settings while increasing it from 0.56 to 0.80 (43% increase) in others, where we note that these estimates relied on clustering to enable standard error estimation (see Methods). This pattern was replicated in the higher-precision, pre-registered validation experiments (8-12 trials per condition), which provided a more reliable representation of heterogeneity, where punishment reduced efficiency from 0.72 to 0.40 (44% decrease) in some settings while increasing it from 0.63 to 0.81 (29% increase) in others (see Figs. 2E and 2F).

The heterogeneity in punishment effects across experimental conditions presents a significant challenge for theory building (*56*). Broad generalizations about punishment effectiveness based on individual PGG settings are inevitably brittle and may not hold even within seemingly similar cooperative contexts. Conversely, developing theories specific to only a single PGG configuration would be of very limited usefulness. A middle ground is to take an integrative approach, identifying the range of conditions over which punishment is likely to enhance or diminish cooperation efficiency. To this end, we frame the problem as an out-of-sample prediction exercise (*37*, *39*, *48*, *49*, *57–59*), where results acquired from one set of experimental conditions can be used to predict punishment effectiveness in entirely new settings. The associated models can, in turn, drive theory development by revealing important relationships between game parameters and punishment effectiveness.

Predicting punishment effectiveness in new experiments

To operationalize the out-of-sample prediction approach, we leveraged the two-wave data-collection process. Specifically, using PGG parameters and the efficiency level in the control condition (punishment disabled), we fit several statistical models—elastic net (E-Net) with two-way interactions, ordinary least squares (OLS), random forest (RF), multilayer perceptron (MLP), and XGBoost (XGB)—to predict efficiency in the treatment condition (punishment enabled) in the Wave 1 learning experiments. We then used these fitted models to predict efficiency in the treatment condition of the Wave 2 experiments (where again the efficiency level in the control condition was a feature in the model) run using 20 new sets of game parameters that were held out for validation. This prediction task reflects the scenario in which a decision-maker observes the status quo (the control condition) and could use the prediction of the outcome under treatment to decide whether to introduce peer punishment. Because decision-makers predict single configurations, we use regular efficiency (net earnings scaled by the full cooperation baseline; pre-registered; see Methods) rather than normalized efficiency, as the former is more intuitive for human judges.

We assessed predictive performance using both root mean squared error (RMSE) and out-of-sample R^2. The latter measures how much better the model's predictions are relative to simply using the average treatment efficiency (i.e., the efficiency of the PGG conditions with punishment available) from the learning experiments (see Model Evaluation in Methods). Effectively, R^2 quantifies the out-of-sample variance explained by the model, or a human judge, compared to this baseline; that is, an R^2 of 0 indicates predictions no more accurate than always predicting the mean treatment efficiency observed in the learning experiments. Correspondingly, R^2 is bounded above by 1 (perfect prediction) but has no lower bound as predictions can be arbitrarily worse than the baseline. To contextualize the predictive power of the models, we invited experts (53 academics: six authors of peer-reviewed PGG papers and 47 forecasters from the Social Science Prediction Platform) and laypeople (500 Prolific workers) to complete the same prediction task. From these individual-level predictions, we derived a collective "wisdom of the crowd" benchmark within each group, defined as the mean prediction across all forecasters.

The E-Net model achieved the best predictive performance ($R^2 = 0.53$, RMSE = 4.52, Fig. 3A), followed by OLS ($R^2 = 0.38$, RMSE = 5.22, Fig. 3B). The collective predictions of both experts and laypeople performed similarly to the baseline ($R^2 = 0.02$, RMSE = 6.53, Fig. 3C; $R^2 = 0.05$, RMSE = 6.44, Fig. 3D, respectively). Strikingly, the superiority of the model over humans applied not only to their averaged predictions but to individuals as well: none of the 553 human judges outperformed the model. This performance gap reveals that human judges struggle to mentally integrate how multiple factors combine to determine outcomes. Notably, experts performed no better than laypeople, suggesting that identifying which factors matter (the focus of most prior work) does not translate into understanding how they combine to produce outcomes.

The E-Net's out-of-sample R^2 of 0.53 compares favorably to recent prediction exercises in the social sciences. While direct comparisons across domains should be interpreted cautiously—and our predictions involve experimental interventions, whereas most prior exercises predict observational outcomes—we note prediction exercises for life-course outcomes (*60*, *61*), Twitter cascade sizes (*62*), and group performance (*63*), for which R^2 varies between 0.05 and 0.4.

Key design factors determining punishment effectiveness.

The models reveal when punishment helps versus harms welfare. Specifically, we assess which features of the best-performing model (E-Net) are most important for predicting punishment effectiveness, using two complementary approaches (See SM Section S9 for analyses for other models). First, we use permutation analysis (*64*) to examine each feature's importance for prediction accuracy by randomly shuffling feature values in the validation experiments and measuring the resulting percentage change in prediction error (Fig. 4A). Features causing larger increases in prediction error when shuffled are more important for prediction. Second, we use SHAP (Shapley Additive Explanations) values (*65*), which quantify how each feature contributes to the model's individual predictions rather than its overall performance (Fig. 4B). These measures are complementary, as permutation feature importance reveals which features are most predictive of punishment effectiveness, while SHAP values show how these features influence the model's predictions. Both are important, as overall feature importance does not necessarily imply effect consistency or vice versa. Here we describe three feature categories: consistently important (communication); important but contingent on interactions with one or more other features (contribution framing, game length); and consistent but weak (MPCR). We do not elaborate on features that are neither predictively important nor consistent in their directionality.

The important (and consistent) role of communication.

Communication has long been recognized as an effective mechanism for promoting cooperation in social dilemmas (*31*, *66–69*); however, its impact on the effectiveness of punishment is unclear (*31*, *70*). Our results show that it is also the dominant factor in determining punishment effectiveness. More precisely, permutation feature importance shows that shuffling this feature increased prediction error by 60%, more than three times that of the next most important feature (Fig. 4A). SHAP values further show that allowing communication has an unambiguously positive impact on punishment effectiveness (Fig. 4B). Given that punishment itself is a public good (i.e., everyone benefits from increased cooperation when free-riders are deterred, but no individual is incentivized to bear the cost of punishing), communication might reinforce social norms by allowing groups to explicitly establish expectations about both cooperation and punishment, potentially reducing arbitrary or retaliatory punishment while ensuring consistent sanctions for non-cooperation (*71*, *72*). It could also enable groups to coordinate sanctioning efforts, preventing both over-punishment of some free-riders and under-punishment of others (*35*, *36*).

Factors that consistently enhance punishment (but vary in importance).

Two other factors showed consistently positive effects on punishment effectiveness, regardless of other game parameters. First, the availability of reward mechanisms increased efficiency in the punishment conditions (Fig. 4B) and was important for prediction accuracy (4.6% increase in error when shuffled; Fig. 4A). Although the relative effectiveness of rewards versus punishment has long been a matter of debate in the literature (*30*, *51*, *73–76*), the results show that the availability of rewards consistently enhanced punishment's effectiveness. Second, higher marginal per capita return (MPCR), which reduces the conflict between individual and collective interests, enhanced punishment effectiveness, though its effect was too small relative to other features to meaningfully impact prediction accuracy.

Factors that are important but have contingent effects.

Several other factors, such as contribution framing (opt-out vs. opt-in), contribution type (variable vs. all-or-nothing), game length, and peer outcome visibility, rank highly in importance but exhibit effects that are contingent on other features. To illustrate, we focus on two of these factors: contribution framing and game length.

First, we find that the contribution framing (opt-out vs. opt-in) strongly influences prediction accuracy (18% increase in prediction error when shuffled). In standard public goods games (opt-in), participants start with their endowment in their private account and must actively choose to contribute to the public good. Under the opt-out framing, this default is reversed: endowments begin in the public good, and participants must actively choose to withdraw funds for private use. Importantly, individuals can only withdraw up to their total endowment, ensuring that both opt-in and opt-out framings are monetarily equivalent. Previous research has shown mixed effects of framing on cooperation (*77–80*). However, to our knowledge, this factor has not been examined in the context of punishment effectiveness. The results reveal a contingent relationship between framing and punishment effectiveness, showing that the impact of opt-out framing depends primarily on whether contributions must be all-or-nothing or can be variable/partial (Fig. 5A and Fig. 5B). The opt-out default enhances punishment effectiveness when participants can make variable contributions but, surprisingly, reduces effectiveness when contributions are restricted to all-or-nothing. This interaction is further modulated by peer outcome visibility: having outcomes be visible amplifies the negative effect of opt-out framing with all-or-nothing contributions but attenuates the positive effect with variable contributions.

Second, game length substantially influences prediction accuracy (11.5% increase in error when shuffled), but its effect on punishment effectiveness depends on two other factors: communication availability and peer outcome visibility. As shown in Figs. 5C and 5D, longer games enhance punishment effectiveness only when communication is available, and this positive interaction is weaker when peer outcomes are visible. These interactions suggest that punishment becomes more effective with communication over repeated iterations of the game because groups may establish and maintain cooperative norms that reduce the need for frequent punishment (*22*, *71*, *72*). The visibility of peer outcomes seems to dampen this positive effect. Although the experiment design cannot identify the mechanism, visibility might reduce punishment effectiveness in multiple ways: participants who observe non-contributors or non-punishers earning higher payoffs may lose motivation to enforce cooperative norms, while the ability to identify and retaliate against punishers could make individuals more hesitant to sanction free-riders (*19*).

Punishment parameters matter less than expected.

In addition to these complex and non-obvious effects, the approach also reveals some surprising null results. For example, Fig. 4A shows that punishment technology, which is defined as the magnitude of a punishment action relative to its cost and hence seems a likely candidate to impact efficiency, had the smallest overall effect on the model's predictive performance of all the features we tested. This unexpected finding could be interpreted in at least two ways. One interpretation suggests that punishment's mechanical design may not be a determinant of its success and that its effectiveness depends more on the context in which it operates (e.g., communication availability, contribution framing, and interaction dynamics). Another explanation reflects methodological limitations, including model (i.e., the model may be capturing idiosyncrasies in the learning data regarding the punishment technology rather than generalizable patterns) and model multiplicity (i.e., many models can achieve similar predictive performance with different structures so a different model might assign greater importance to punishment parameters).

Discussion

The standard approach for social and behavioral experiments is to study factors in isolation, holding "all else constant" to identify causal effects. This paradigm has generated substantial knowledge but faces a fundamental limitation: when multiple factors influence outcomes and interact with each other, studying them individually cannot reveal how they combine to determine results. The integrative experiment demonstrates what becomes visible when this constraint is relaxed. We systematically varied 14 design parameters across 360 experimental conditions and found that punishment's effect on cooperation efficiency ranges from +43% to -44% depending on game parameters. This heterogeneity follows predictable patterns, as models trained on the experiments successfully predicted outcomes in entirely new contexts.

Three substantive findings illustrate what this approach reveals. First, communication is roughly three times more important than any other parameter in predicting punishment effectiveness. This quantification of relative, out-of-sample predictive importance for punishment's effect across a broad parameter space represents a different type of knowledge than simply knowing that communication "helps" cooperation or welfare. Second, contribution framing (opt-in vs opt-out) emerged as the second most important predictor. The importance of this factor is particularly striking given how little attention it has received in the literature. Third, interactions reveal boundary conditions for when factors matter. For example, game length consistently enhances punishment's effectiveness only when communication is available, whereas contribution framing effects depend on both contribution type and outcome visibility. Such interactions and their boundary conditions are inherently invisible when holding "all else constant."

The present study has important limitations that clarify its scope. Public goods games are stylized representations that capture the essential tension between individual and collective interests in a controlled setting. This stylization enables systematic investigation but limits external validity. Whether our specific findings about punishment effectiveness extend to other cooperation games or real-world settings must be clarified in future work. Additionally, punishment effectiveness varies across populations (*18*, *20*, *52*), but the study deliberately isolated variation in game parameters to establish that such factors are consequential even within homogeneous populations. That we find such substantial heterogeneity within the most studied population suggests that interactions between population and game parameters could reveal even more complex patterns, and that the prediction exercise likely represents an optimistic assessment of performance, for both quantitative models and human forecasters.

Importantly, the integrative approach itself generalizes beyond punishment in public goods games. Many phenomena in social science exhibit context (and population) sensitivity that produces mixed and seemingly contradictory findings across studies (*37*, *81*, *82*). The integrative approach addresses this by shifting the scientific question from "does this effect exist?" to "how much does it matter relative to other effects, and when?" This reframing is what enables prediction of outcomes when multiple factors operate simultaneously, which is the fundamental question single-factor designs cannot answer.

Implementing this approach requires several design choices researchers can adapt to their goals. The 14-dimensional design space used in this study captures several key aspects of cooperative settings, but it is not intended to be comprehensive or definitive. Rather, the contribution should be viewed as the first step in an iterative process of design space construction and empirical testing (*38*). Design parameters serve as features whose utility can be empirically evaluated through out-of-sample prediction. Those that fail to predict punishment effectiveness may be removed or refined, while predictively important factors will be retained and potentially disaggregated into finer-grained constituent features (e.g., "communication" could be further decomposed into, for instance, its semantic or structural components). In this way, unexpected findings can guide the modification of existing dimensions or suggest new ones, allowing the framework to evolve based on clear empirical criteria (i.e., its ability to predict punishment effectiveness in new settings) rather than relying solely on theoretical arguments or statistical relationships in existing data.

The breadth and structure of the dataset represents another contribution. Unlike datasets generated by testing specific hypotheses, the data emerged from space-filling parameter exploration, making it particularly valuable for informing and constraining theory development as well as training and validating predictive models. Again, however, the sampling strategy represents another design choice that can be adapted to research goals. We used Sobol sequences and random sampling for uniform coverage of the parameter space, but researchers could instead employ active learning methods that sequentially select the most informative conditions (*83*–*86*), which can substantially increase efficiency when experiments are costly or time-consuming. As for the number of trials per condition: we used single trials for 320 learning experiments to maximize breadth, then 8-12 trials for 20 validation experiments to enable precise comparison with human forecasters. Researchers without this constraint could sample more conditions with single trials, averaging predictive performance across conditions to maximize parameter space coverage under fixed budgets (*44*–*47*). The choice between random versus adaptive sampling, and between breadth versus precision, depends on whether the goal is comprehensive parameter space coverage or focused investigation of specific phenomena identified as theoretically or practically important.

Model interpretation represents another consideration. The interpretability methods we employ serve to make the model's learned patterns more understandable to humans (*87*). The goal of such interpretation is twofold: it can reveal something interesting about the studied phenomenon (i.e., a new discovery, such as the dominance of communication or previously unhypothesized interaction), or it can help improve the model's performance under novel or shifting conditions. However, these methods necessarily simplify the model's complexity to enhance cognitive accessibility, which means they should be viewed as aids to understanding rather than complete representations of what the model learned. For a broader overview of interpretability research in machine learning, see Molnar (*88*) and Lipton (*89*); for interpretability specifically within scientific contexts, see Molnar and Freiesleben (*90*).

Many phenomena in social science are shaped by a large number of factors whose interactions are consequential. Yet the dominant paradigm often asks only "does this effect exist?" and examines hypothesized factors in isolation. As a result, research programs can accumulate many partial explanations without a clear picture of how they combine to determine outcomes. Knowing that factors matter is fundamentally different from knowing how much each matters and how they interact. The integrative approach implemented here offers one way forward. It varies many factors within a shared design space, evaluates models by their accuracy on new experiments, and probes those models to constrain and develop theory. Our hope is that integrative experiments, combined with models that integrate prediction and explanation, represent a path toward more cumulative social science

Materials and methods summary

We recruited 7,100 participants who were randomly assigned to play synchronous public goods games (PGGs) with systematically varied design parameters. Participants were recruited from two online platforms: Amazon Mechanical Turk (via a curated panel) and Prolific. The experiments were implemented using Empirica (50), an open-source software for integrative experimentation through multi-participant online games with real-time interactivity. For screenshots of the interface and its implementation details, refer to SM Section S4. The experiments were executed in two waves. In Wave 1, 3,618 participants played 335 PGGs with configurations determined by a Sobol sequence to achieve uniform coverage of the 14-dimensional design space. Consistent with recent related methodological innovations and experimental design principles showing that, for a fixed overall sample size (i.e., fixed budget), estimation accuracy for models with interactions is maximized by sampling many unique design points rather than repeated observations (44–47), each experiment was conducted once. This data collection wave provides a dataset with high breadth but low precision across the parameter space, allowing us to explore and uncover empirical patterns between cooperative settings and outcomes. In Wave 2, 3,482 participants played 417 PGGs with 40 different configurations selected by random sampling. Each PGG consisted of multiple rounds with contribution, redistribution (where punishment/reward could occur), and round summary stages. Throughout the study, 14 design parameters were manipulated, including group size, game length, communication availability, punishment and reward mechanisms, and the marginal per capita return. For complete details, see SM Sections S2-S4.

We measured group cooperation using two related metrics. First, we calculated conventional efficiency as the ratio of the group's total earnings to the earnings of a hypothetical fully cooperative group:

$$Efficiency = \frac{E_{group}}{E_{full\ cooperation}}$$

where E_{group} is the group's total earnings and $E_{full\ cooperation}$ represents earnings if all members contributed fully and never punished or rewarded. This conventional measure was used in the human forecasting task, as it is familiar to expert forecasters and appropriate for comparing treatment and control conditions with the same configuration. To compare cooperative behavior across different PGG configurations, we introduce normalized efficiency that accounts for differences in the range of possible efficiencies:

$$Normalized\ Efficiency\ =\ \frac{E_{group} - E_{full\ defection}}{E_{full\ cooperation} - E_{full\ defection}},$$

where $E_{full\ defection}$ represents earnings if all members contributed nothing and never punished or rewarded. This measure enables direct comparison between PGGs of different designs.

We tested heterogeneity in punishment effectiveness using Cochran's Q-statistic, the Fisherian randomization test, and I^2 (SM Section S7). We trained five statistical models to predict average group efficiency under punishment: ordinary least squares, elastic net, random forest, XGBoost, and multilayer perceptron. Models were trained on Wave 1 data and evaluated on held-out Wave 2 data (SM Section S8). To contextualize model performance, we compared predictions to those of 53 experts and 500 laypeople (SM Section S10). Comparisons of predictive performance are assessed using paired bootstrap differences computed on the same resamples; for each comparison, we report the mean difference in RMSE and the 95% CI derived directly from the empirical distribution of the differences (SM Section S8). Feature importance was assessed using permutation feature importance and Shapley additive explanations (SM Section S9). The study was determined exempt by MIT's IRB (Protocol #E-5462). Wave 2 configurations and the human forecasting task were pre-registered (AsPredicted #182022, #191059). All other data and analyses in this work are exploratory. For deviations from the pre-registration and associated sensitivity analyses, see SM Section S1.

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Supplementary Materials

Materials and Methods

Supplementary Text

Figs. S1 to S14

Tables S1 to S8

References (*92–104*)

Fig. 1. Experimental design and implementation. Panel (A) depicts a schematic representation of a typical public goods game, illustrating the basic tension between individual and collective

interests. Panel (B) shows how each PGG setting is defined by a 14-dimensional vector of design parameters, with pairs of games—treatment (punishment enabled) and control (punishment disabled)—sharing otherwise identical design parameters. Data collection occurred in two waves: Wave 1 (learning experiments) used a Sobol sequence to systematically sample 320 experimental conditions with one realization each, while Wave 2 (validation experiments) randomly selected 40 new experimental conditions with multiple realizations each, providing higher precision but lower breadth. Panel (C) shows the interface from a participant's perspective, progressing through contribution, distribution, and summary stages. Interface elements (e.g., chat functionality, outcome visibility, punishment/reward options) correspond to varying PGG design parameters across games. See Methods for details.

Fig. 2. Average punishment effects and heterogeneity across conditions. Panel (A) shows the average contribution per round (as a percentage of per round endowment) to the public good in games with and without punishment. Panel (B) shows normalized efficiency in games with and without punishment. Panel (C) shows normalized efficiency with and without punishment (filled markers: means; open markers: individual games); Panel (D) shows punishment effects. Panels (E) and (F) show the corresponding analyses for validation experiments, where multiple trials per condition enabled direct estimation without clustering. The dashed line in Panels (C) and (E) represents the mean normalized efficiency in the learning and validation experiments respectively. Points outside the y-axis range are shown at the boundary with red outlines and labeled with their actual values. In the learning experiments, we grouped conditions by their design parameters into 20 clusters to enable standard error estimation. Error bars show 95% confidence intervals. See Methods for clustering details.

Fig. 3. Predictive performance on unseen experiments. Panels (A-D) show the predicted versus true efficiency values for the 20 validation experiments. The diagonal line represents perfect prediction. The dashed line shows the baseline prediction, while the regions shaded in green indicate where predictions have lower absolute error than the baseline. Each panel includes the out-of-sample R^2 and root mean squared error (RMSE) of the predictions. All predictions were made using the PGG design parameters and efficiency in the control condition (punishment disabled) to predict efficiency in the treatment condition (punishment enabled). The observed performance metrics in Panels (A-D) are descriptive quantities for the specific validation set, but one can also view these experiments as a random sample from the population of all possible PGG configurations. Panel (E) shows bootstrapped 95% confidence intervals for RMSE across prediction approaches. The E-Net model outperformed both experts (95% CI of difference in RMSE: [-0.10, 3.71]) and laypeople (95% CI of difference in RMSE: [0.03, 3.86]), while experts and laypeople showed no significant difference in performance (95% CI of difference in RMSE: [-2.4, 2.6]). See Methods for additional details.

Fig. 4. Feature importance and model interpretation. Panel (A) shows each feature's importance to out-of-sample prediction. For each design parameter, the feature's importance to the E-Net model's predictive performance is measured as the ratio of the model's error in predicting the outcome of validation experiments when the values for that feature are randomly shuffled, and its baseline error on the original, unpermuted set of validation experiments. Each feature is shuffled independently, and error bars indicate 95% confidence intervals arising from the random parameter shuffles. Panel (B) displays SHAP values showing how each feature contributes to individual predictions. Each point represents one learning experiment, with color indicating feature value (e.g., for communication: blue = disabled, pink = enabled) and

horizontal position showing the magnitude and direction of the contribution to predicted efficiency. Features whose points appear primarily on the right increase predicted efficiency, while those on the left decrease it. The horizontal spread reflects how a feature's impact varies across experiments. For instance, while communication generally increases efficiency when enabled (pink dots mostly on the right), its magnitude depends on other design parameters in each specific game configuration. Note: The model also uses the control (no punishment) efficiency to inform predictions, although it is omitted from the figure for clarity, as it was not a design parameter. Of the 14 design parameters, the model uses 12: reward technology (and cost) are excluded as their values are only meaningful when rewards are enabled. See SM Section S9 for similar analyses for the other models.

Fig. 5. Key interaction effects shaping punishment effectiveness. Panels (A) and (B) show the interaction between contribution framing and contribution type. The opt-out default enhances punishment effectiveness with variable contributions but reduces it with all-or-nothing contributions. The effect is amplified for all-or-nothing contributions and attenuated for variable contributions when peer outcomes are visible. Panels (C) and (D) show how game length interacts with communication and peer outcome visibility. Longer interactions enhance punishment effectiveness only when communication is available, and this positive effect is weaker when peer outcomes are visible.

Table 1. Design space parameters

Design Parameter	Description	Values
Group Size	Number of players in the game	2-20 players
Game Length	Number of rounds in the game	1-30 rounds
Contribution Type	Whether players can contribute any amount or must choose between contributing all or none of their endowment	{variable, all-or-nothing}
Contribution Framing	Whether each player's endowment starts in their private account (opt-in) or in the public fund (opt-out)	{opt-in, opt-out}
MPCR	Marginal per capita return on contributions, defined as the fund multiplier divided by group size	0.06-0.7
Communication	Whether players can communicate through a chat window during the game	{enabled, disabled}
Peer Outcome Visibility	Whether players can see summaries of others' earnings and punishments/rewards received in each round	{visible, hidden}
Actor Anonymity	Whether the identity of players who punish or reward others is revealed	{revealed, hidden}

Design Parameter	Description	Values
Horizon Knowledge	Whether players are shown the total number of rounds and rounds remaining	{known, unknown}
Punishment	Whether players can impose costly penalties on others; this is the focal treatment	{enabled, disabled}
Peer Incentive Cost	Number of coins a player must spend to impose one unit of punishment or grant one unit of reward	1-4 coins
Punishment Technology	Number of coins deducted from the punished player per coin spent punishing	1-4 coins
Reward	Whether players can grant costly rewards to others	{enabled, disabled}
Reward Technology	Number of coins granted to the rewarded player per coin spent rewarding	0.5-1.5 coins